



Quantitative Multi-Asset Financial Intelligence & Backtesting Platform

Problem Statement

Financial markets generate large volumes of historical data across different asset classes such as **commodities, cryptocurrencies, and equities**. However, investors and financial researchers often depend on multiple tools to analyze trends, measure risk, compare asset performance, and test quantitative strategies.

The proposed system aims to develop a **Quantitative Multi-Asset Financial Analysis Platform** capable of automatically collecting, processing, and analyzing historical market data for assets such as **Gold, Bitcoin, and NVIDIA**.

The platform should apply quantitative and statistical techniques including **SMA, EMA, returns analysis, volatility, Sharpe ratio, maximum drawdown, correlation, and rolling performance analysis** to understand how different assets behave under changing market conditions.

A major component of the platform will be a **strategy backtesting engine** that allows users to simulate predefined quantitative strategies using historical data while considering factors such as **initial capital, transaction costs, entry/exit signals, position sizing, and benchmark performance**.

The goal is to create a practical **Quantitative Finance and FinTech research platform** that transforms raw market data into structured, meaningful, and visually understandable financial insights.

Core Innovations

The proposed solution integrates **Financial Data Engineering, Quantitative Analysis, Risk Modelling, Strategy Backtesting, and Interactive Visualization** into a unified platform.

Rather than functioning only as a market-price dashboard, the system should help users understand **asset behaviour, changing correlations, historical risk, strategy performance, and market conditions**.

Key Features

- **Multi-Asset Data Processing**

Collect and normalize historical data for **Gold, Bitcoin, NVIDIA**, and optionally other financial assets.

• **Quantitative Indicator Engine**

Calculate indicators such as:

- SMA and EMA
- Daily and cumulative returns
- Historical and annualized volatility
- Sharpe Ratio
- Maximum Drawdown
- Rolling returns

• **Cross-Asset Correlation Analysis**

Generate a **correlation matrix** and rolling correlation analysis to study how relationships between different assets change over time.

• **Strategy Backtesting Engine**

Allow users to test strategies such as:

- SMA Crossover
- EMA Trend Strategy
- Momentum Strategy
- Mean Reversion Strategy

Backtesting should simulate portfolio performance instead of simply displaying buy and sell signals.

• **Realistic Trading Simulation**

The backtesting engine should consider:

- Initial capital
- Position sizing
- Transaction costs
- Entry and exit prices
- Portfolio value
- Number of trades

• **Strategy vs Benchmark Comparison**

Compare each strategy against a **Buy-and-Hold benchmark** using metrics such as return, Sharpe ratio, volatility, and maximum drawdown.

• **Strategy Robustness Testing**

Allow users to modify parameters such as moving-average periods, transaction costs, and backtesting periods to understand whether a strategy remains effective under different conditions.

• **Market Regime Analysis**

Analyze strategy behaviour during different market environments such as:

Bull Market | Bear Market | High Volatility | Low Volatility

• **Interactive Financial Dashboard**

The platform should visualize:

- Price trends
- SMA/EMA indicators
- Returns and volatility
- Drawdowns
- Correlation heatmaps
- Buy/Sell signals
- Backtesting equity curves
- Strategy vs benchmark performance

Financial Considerations

The platform is intended for **quantitative research and historical analysis** and should not present historical strategy performance as guaranteed future returns.

The backtesting system should minimize issues such as **look-ahead bias, data leakage, unrealistic trade execution, and over-optimization**.

Future Scope

Future improvements may include **portfolio optimization, Monte Carlo simulation, Value at Risk, machine-learning-based market regime detection, paper trading, real-time market data, and AI-powered quantitative research assistance**.

Conclusion

The **Quantitative Multi-Asset Financial Intelligence & Backtesting Platform** aims to provide a unified environment for analyzing multiple financial assets, measuring risk, studying correlations, and evaluating quantitative strategies.